

VAIBHAV KUMAR KAUSHAL

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EDUCATION

Thapar Institute of Engineering & Technology (TIET) | Patiala, India *Bachelor of Engineering (B.E.) in Computer Engineering (COE)*

- Expected Graduation: 2030
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks.

TECHNICAL SKILLS

- **Languages:** C (C11/C17), Bash Shell Scripting, Python (Basic), C++
- **Systems & OS:** Kali Linux, Ubuntu/Debian, POSIX API, Memory Management, Process Automation
- **Cybersecurity:** TCP/IP Protocols, Penetration Testing Utility Familiarity (Kali Stack), Network Reconnaissance, Vulnerability Auditing, Privilege Escalation Analysis
- **Developer Tools:** Git/GitHub, GCC Compiler, GDB Debugger, Makefiles, Wireshark, Nmap

ENGINEERING PROJECTS

AutoRecon — Multi-Threaded Security Reconnaissance Framework | C, Bash, POSIX Threads, Kali Linux Tooling

- Engineered a lightweight network reconnaissance tool in low-level C utilizing raw socket programming to scan target networks.
- Implemented multi-threading using pthreads to execute concurrent port scanning, reducing scan times significantly compared to sequential execution.
- Wrapped the C executable in a robust Bash automation engine that parses IP inputs, coordinates execution, and automatically formats output logs for immediate security analysis.

SysHardener — Linux Security & Privilege Auditor | Bash, Linux Core Utilities, System Administration

- Developed an automated Bash shell script designed to evaluate the security posture and configuration of Linux distributions.
- Scripted conditional checks to detect excessive world-writable permissions, audit exposed open ports via netstat/ss, and highlight suspicious SUID/GUID binaries.
- Designed the utility to output a clean, actionable terminal report detailing potential vulnerabilities and recommended system hardening steps for blue-team operations.

Interactive Terminal Portfolio | *HTML5, CSS3, JavaScript*

- Built a custom, fully interactive web terminal utilizing raw web technologies to showcase systems programming projects.
- Integrated a dynamic canvas-based rendering engine to handle background animations without sacrificing browser performance.
- Developed a custom JavaScript command parser to allow users to navigate the portfolio using standard Linux CLI commands.

EXTRACURRICULAR & COMPETITIVE ACTIVITIES

- **CTF Player & Security Enthusiast:** Actively practice binary exploitation, reverse engineering, and network packet analysis via platforms like PicoCTF and TryHackMe.
- **Linux Environment Customization:** Daily driver of Linux environments, continually writing custom shell scripts to automate local development workflows and system monitoring.